

Introduction

1.1. Definitions.

Differential Equation : Equations such as

$$(i) \quad \frac{dy}{dx} = \frac{\sqrt{1-x^2}}{\sqrt{2-y^2}},$$

$$(ii) \quad \left(\frac{dy}{dx}\right)^2 + 2y^2 = 4 \left(\frac{dy}{dx}\right) + 4x,$$

$$(iii) \quad \frac{d^3y}{dx^3} + 7 \frac{d^2y}{dx^2} + 8 \frac{dy}{dx} - 9y = \log x,$$

$$(iv) \quad \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = kz,$$

$$(v) \quad \frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0,$$

which involve differential coefficients, are called the *differential equations*.

Ordinary Differential Equations. Equations like (i), (ii), (iii) which involve a single independent variable are called *ordinary differential equations*.

Partial Differential Equations. Equations like (iv) and (v) which involve partial differential coefficients with respect to more than one independent variable are called *partial differential equations*.

Order and Degree of Differential Equations. An equation like (iii) which involves a third order differential coefficient but none of higher order is said to be of third order. Equation (v) is also of second order.

The degree of a differential equation is the power (or degree) of the highest differential coefficient when the equation has been made rational. Thus equations (i), (iii), (iv) and (v) are all of first degree and equation (ii) is of second degree.

General Solution. The relation containing n arbitrary constants which satisfies an ordinary differential equation of n th order is called its *complete primitive* or *general solution*.

It can be shown that by eliminating n arbitrary constants from an equation in x, y , we get a differential equation of n th order. Such a process is called *formation of differential equations*.

Particular Solution. A particular solution of differential equ-

ation is one obtained from the primitive by assigning definite values to the arbitrary constants.

Geometrically, the primitive is the equation of a family of curves satisfying the differential equation and a particular solution is the equation of some one of this family of curves.

We now give examples on the formation of differential equations by eliminating the arbitrary constants.

Ex. 1. Eliminate the constants from $y = ax + bx^2$. [Nag. 61 (S)]

Solution. We have $\frac{dy}{dx} = a + 2bx$, $\frac{d^2y}{dx^2} = 2b$.

$$\text{Now } a = \frac{dy}{dx} - 2bx = \frac{dy}{dx} - x \frac{d^2y}{dx^2}, \quad b = \frac{1}{2} \frac{d^2y}{dx^2}.$$

Putting these values of the constants in given equation, we get

$$y = x \left(\frac{dy}{dx} - x \frac{d^2y}{dx^2} \right) + \frac{1}{2} x^2 \frac{d^2y}{dx^2}$$

$$\text{or } x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0,$$

which is a differential equation of second order, obtained from $y = ax + bx^2$ after eliminating the arbitrary constants a and b .

Ex. 2. Eliminate the constant a from

$$\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y). \quad \dots(1)$$

Solution. Differentiating the given equation, we get

$$\frac{-x}{\sqrt{1-x^2}} - \frac{y}{\sqrt{1-y^2}} \frac{dy}{dx} = a \left(1 - \frac{dy}{dx} \right). \quad \dots(2)$$

Dividing (1) and (2) to eliminate a , we get

$$-\frac{\sqrt{1-x^2} + \sqrt{1-y^2}}{x/\sqrt{1-x^2} + y/\sqrt{1-y^2} (dy/dx)} = \frac{x-y}{(1-dy/dx)}$$

This can be simplified to give $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$.

We find here that after eliminating one arbitrary constant a , we get a differential equation of first order.

Ex. 3. Form the differential equation of which

$$c(y+c)^2 = x^3, \quad \dots(1)$$

is the complete integral.

Solution. Differentiating the given equation, we get

$$2c(y+c) \frac{dy}{dx} = 3x^2. \quad \dots(2)$$

Dividing (1) by (2),

$$\frac{y+c}{2 dy/dx} = \frac{x}{3}, \text{ i.e., } 3(y+c) = 2x \frac{dy}{dx}$$

$$\text{or } c = \frac{1}{3} \left(2x \frac{dy}{dx} - 3y \right).$$

Putting this value of c in (2), we get

$$\frac{2}{3} \left(2x \frac{dy}{dx} - 3y \right) \cdot \frac{2}{3} x \frac{dy}{dx} = 3x^2,$$

$$\text{i.e., } 8x \left(\frac{dy}{dx} \right)^2 - 12y \left(\frac{dy}{dx} \right) = 27x,$$

a differential equation of first order and third degree.

Ex. 4. Form the differential equation corresponding to the family of curves $y = c(x - c)^2$ where c is an arbitrary constant.

[Karnatak 1960]

Solution. Here $\frac{dy}{dx} = 2c(x - c)$.

Dividing the given equation by (1),

$$\frac{y}{\frac{dy}{dx}} = \frac{x - c}{2} \quad \text{or } c = x - \frac{2y}{p}, \quad \text{where } p = \frac{dy}{dx}.$$

Putting this value of c in (1), we get

$$p = 2 \left(x - \frac{2y}{p} \right) \left(\frac{2y}{p} \right), \quad \text{i.e., } p^3 = 4y(px - 2y),$$

which is the required differential equation.

Ex. 5. Find the differential equation of all circles passing through the origin and having their centres on the x -axis.

[Nag. T.D.C. 1961]

Solution. Equations of circles passing through the origin and having their centres on the x -axis is

$$x^2 + y^2 + 2gx = 0,$$

where g is an arbitrary constant.

$$\text{Differentiating, } x + y \frac{dy}{dx} + g = 0, \quad \text{i.e., } g = - \left(x + y \frac{dy}{dx} \right).$$

Putting this value of g in the equation of circles, we get

$$x^2 + y^2 - 2x \left(x + y \frac{dy}{dx} \right) = 0, \quad \text{i.e., } y^2 = x^2 + 2xy \frac{dy}{dx}$$

which is the required differential equation.

Ex. 5. Find the differential equation of the family of parabolas with foci at the origin and axis along the x -axis.

Solution. Let the directrix be $x = -2a$ and latus rectum be $4a$. Then equation of the parabola is

(distance from focus = distance from directrix),

$$x^2 + y^2 = (2a + x)^2 \quad \text{or } y^2 = 4a(a + x). \quad \dots(1)$$

$$\text{Differentiating, } y \left(\frac{dy}{dx} \right) = 2a, \quad \text{or } a = \frac{1}{2} y \frac{dy}{dx}.$$

Putting this value of a in (1), the differential equation is

$$y^2 = 2y \frac{dy}{dx} \left(\frac{1}{2} y \frac{dy}{dx} + x \right) \quad \text{or } y \left(\frac{dy}{dx} \right)^2 + 2x \left(\frac{dy}{dx} \right) - y = 0.$$

Ex. 7. Form the differential equation that represents all parabolas each of which has a latus rectum $4a$ and whose axes are parallel to x -axis.

Solution. Equation of the family of such parabolas is

$$(y-k)^2 = 4a(x-h), \quad \dots(1)$$

where h and k are arbitrary constant.

$$\text{Differentiating, } (y-k) \frac{dy}{dx} = 2a. \quad \dots(2)$$

$$\text{Differentiating again, } (y-k) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0. \quad \dots(3)$$

Putting value of $y-k$ from (2) in (3), we get

$$2a \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0,$$

which is the required differential equation.

Ex. 9. Form the differential equation of all parabolas whose axes are parallel to the axis of y .

Solution. Such parabolas are given by

$$(x-h)^2 = 4a(y-k),$$

where h, k, a are three arbitrary constants.

$$\text{Differentiating, } (x-h) = 2a \frac{dy}{dx}.$$

$$\text{Differentiating again, } 1 = 2a \frac{d^2y}{dx^2} \quad \text{i.e., } \frac{d^2y}{dx^2} = \frac{1}{2a}.$$

$$\text{Differentiating once again, } \frac{d^3y}{dx^3} = 0.$$

This is the required differential equation.

Ex. 9. Form differential equation of all conics whose axes coincide with the axes of co-ordinates. [Delhi Hons. 1958]

Solution. Such conics are given by

$$ax^2 + by^2 = 1, \quad \dots(1)$$

where a and b are two arbitrary constants.

$$\text{Differentiating, } \frac{dy}{dx} = -\frac{ax}{by}, \quad \frac{d^2y}{dx^2} = -\frac{a}{b} \left(\frac{1}{y} - \frac{x}{y^2} \frac{dy}{dx} \right),$$

$$\text{i.e. } \frac{d^2y}{dx^2} = \frac{y}{x} \frac{dy}{dx} \left(\frac{1}{y} - \frac{x}{y^2} \frac{dy}{dx} \right) \quad \text{as } -\frac{a}{b} = \frac{y}{x} \frac{dy}{dx}$$

$$\text{or } xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx},$$

which is the required differential equation.

Ex. 10. Form the differential equation in the following cases :

(i) $y = c \sin^{-1} x$ (c parameter).

$$\text{Ans. } y = \sqrt{(1-x^2)} \frac{dy}{dx}$$

(ii) $y = ae^{2x} + be^{-2x} + ce^x$ (a, b, c parameters).

Ans. $\frac{d^2y}{dx^2} - 7 \frac{dy}{dx} + 6y = 0.$

(iii) $ay^2 = (x-c)^3$ (c parameter).

Ans. $8a \left(\frac{dy}{dx} \right)^3 = 27y.$

(iv) $y = cx + c - c^2$ (c parameter).

Ans. $y = x \frac{dy}{dx} + \frac{dy}{dx} - \left(\frac{dy}{dx} \right)^2.$

(v) $e^{2y} + 2cxe^y + c^2 = 0$ (c parameter).

Ans. $(1-x^2) \left(\frac{dy}{dx} \right)^2 + 1 = 0.$

(vi) $y = a \cos (mx+b)$ (a, b parameters).

Ans. $\frac{d^2y}{dx^2} + m^2y = 0.$

(vii) $xy = ae^x + be^{-x}$ (a, b parameters).

Ans. $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - xy = 0.$

/ (viii) $xy = Ac^x + Be^{-x} + x^2.$

[Osmania 60]

Ans. $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = (xy - x^2) + 2$

Ex. 11. Find the differential equation of all circles of radius a .

[Delhi Hons. 66; Poona 63]

Hint. Equation of the circle is $(x-h)^2 + (y-k)^2 = a^2$.

Eliminate h and k to get the diff. equation in the usual way.

Ex. 11. (b) Find the differential equation of all circles which have their centres on x -axis and have a given radius.

[Marathwada 60]

Hint. Equation of the circle is $(x-h)^2 + y^2 = a^2$, where h is the parameter.

Ex. 12. Define (i) General solution, (ii) Particular solution of a differential equation and obtain the differential equation of the family of curves $y = e^x (A \cos x + B \sin x)$.

[Poona 64]

Hint. Curve is $y = e^x (A \cos x + B \sin x)$.

$$\begin{aligned} \therefore \frac{dy}{dx} &= e^x (A \cos x + B \sin x) + e^x (-A \sin x + B \cos x) \\ &= y + e^x (-A \sin x + B \cos x). \end{aligned}$$

$$\begin{aligned} \text{Also } \frac{d^2y}{dx^2} &= \frac{dy}{dx} + e^x (-A \sin x + B \cos x) - e^x (A \cos x + B \sin x) \\ &= \frac{dy}{dx} + \left(\frac{dy}{dx} - y \right) - y. \end{aligned}$$

$$\text{Thus } \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$$

is the required differential equation.